HS4002: Lecture 9B

Dimensionality Reduction I

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Using AI Agents through Command Line Interfaces

Agents through command line

- The terminal is a program that takes text commands. On a Mac it is called Terminal. On Windows, use Git Bash or PowerShell.
- You have already used it. `cd` changes folder. `git status` asks git what changed.
- We're going to learn to use agentic AI tools through the CLI.
- The folder you are in matters. A command acts on that folder, not on "your computer" in the abstract.

A chat window and a harness

- A chatbot in a browser can only talk. You copy the code out, paste it into a notebook, and run it yourself.
- A harness is the program wrapped around the model. It lets the model read files, edit them, and run commands, then look at what happened.
- We will use **Claude Code**. It is that harness, opened from the terminal, in a folder.

Why the folder matters

- Claude Code can see the files in the folder you opened, and the folders inside it.
- It cannot see a file that lives somewhere else, and it should not be pointed at your whole home folder.
- Open it in the lab folder, or in the team repository. “Clean this up” then means that project, and not your Desktop.

Install it once

Claude Code is a paid subscription, like Cursor.

```
# macOS or Linux
curl -fsSL https://claude.ai/install.sh | bash

# Windows, in PowerShell
irm https://claude.ai/install.ps1 | iex
```

Close the terminal, open a new one, and check:

```
claude --version
```

The first time you run `claude`, it asks you to sign in.

Start it in the right folder

```
cd ~/Documents/teaching/hs4002/ (THIS IS AN EXAMPLE -- DO WHATS APPROPRIATE FOR YOU)
claude
```

- You get a prompt. Type what you want in ordinary English.
- `/help` lists the commands. `/clear` starts a fresh conversation. `/exit` quits.
- Leave it running while you work. It remembers this conversation until you clear it.

What one turn looks like

1. You describe the task.
2. Claude Code reads the relevant files and proposes an edit or a command.
3. You accept or refuse.
4. It runs the command, reads the output, and continues until the task is done or it gets stuck.

It will stop and show you a command before it runs one. That pause is the point.

Plan before it edits

- Press Shift+Tab until the mode says **plan**. In plan mode it reads and proposes. It does not edit yet.
- Read the plan the way you would read a classmate's proposal. If the steps are not what you meant, say so.
- Shift+Tab again when you are ready for it to make the edits.
- If a turn is going somewhere you did not ask for, stop it and restate the task.

Ask for something you can check

A useful ask names what to read, what to write, and what not to touch.

```
Read the lab notebooks from previous weeks. Do not edit them,
and do not change any other file.
Write review.md, and nothing else. For each notebook, say
what it does, which variables it uses, and anything that looks off.
```

"Do the lab" is not an ask. You will not be able to tell a finished notebook from a plausible one.

Read the command before you accept it

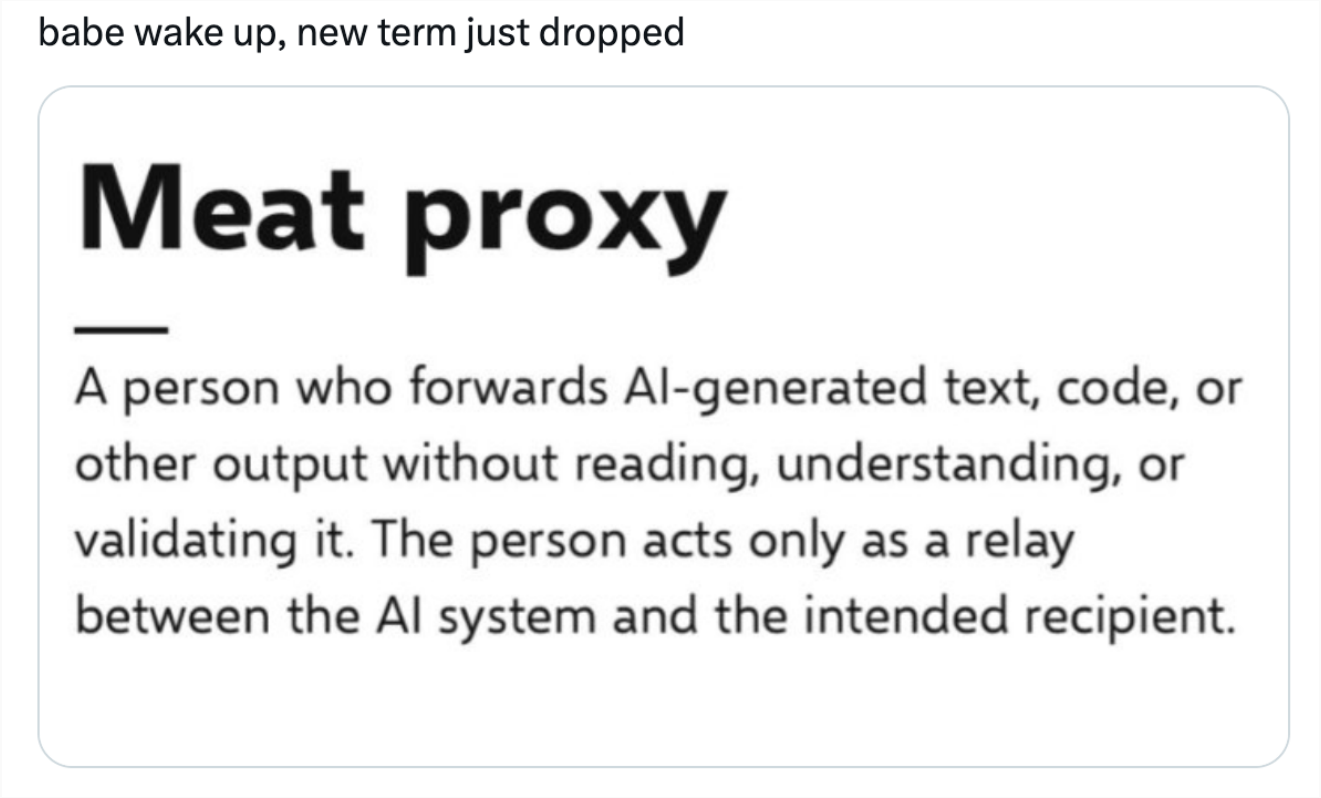
- A command you understand, such as `python week9_notebook.py`, is fine to accept.
- Stop if you see `rm`, `git push`, or anything that mentions a password, a token, or a key.
- If you do not know what a command does, ask Claude Code to explain it before you accept. Then decide.
- Accepting is you running the command. The model does not take the blame for it.

Leave a note in the folder

A file named `CLAUDE.md`, in the project folder, is read at the start of every conversation. Keep it short.

```
This is an HS4002 lab.
Use Python, pandas, scikit-learn, and plotnine.
Do not add packages we have not used.
Explain a change in a sentence before you make it.
```

Do not be meat proxy



What you do not hand over

- Authorship! You must stay in 'control' – you *lead* Claude (in your own limited ways), not the other way round.
- You are accountable for everything. I never want to hear about the choices Claude/Codex/DeepSeek made for you.
- The same rule as week 1: do not rely on it for a result you could not debug, or could not explain, yourself.
- Never delegate writing to AI.
- Never delegate revision to AI in an unsupervised way.

Introduction to PCA

Exercise

- National exams like the GCE O or A levels involve the assessment of student competency across multiple subject areas.
- How do organizations/institutions (like MOE or universities admissions) compare and evaluate students currently?
- What are some problems with current approaches? How might you imagine an alternative?

Principal Components Analysis

Principal components analysis is a dimensionality reduction technique that uses orthogonal linear transformations of data to reduce them to lower-dimensional representations (the “principal components").

Why PCA?

PCA is a popular technique both in the social sciences and data science:

1. It's a way of reducing the dimensionality of the data (often we can use it in regressions to avoid overfitting).
2. The latent dimensions they uncover can themselves be objects of study.
3. It lets us project high dimensional data into lower (e.g. 2) dimensional graphs.

Intuition behind PCA

Imagine the data as a matrix (math for “table") where each row is an individual, and each column is - some - variable (e.g. income, age).

PCA is a way of representing this space/table using fewer dimensions than before, while disrupting the distances between each person as little as possible.

What is a Principal Component?

- PCs refer to particular linear combinations of the different variables included in the data matrix.
- You can think of each PC as an "axis" or "dimension."
- They represent the directions of maximum variability and provided a simple and more parsimonious description of the covariance structure of the data.

What is a Principal Component?

- Each PC describes a different amount of variance, in descending order.
  - $\rightarrow$ The 1st PC describes the most, followed by the 2nd PC and so on ...

Math Behind PCA

Math Behind PCs

The first principal component of a set of variables $X_1, X_2 ... X_p$ is the normalized linear combination of the variables

$$Z_1 = \phi_{11} X_1 + \phi_{21} X_2 + ... \phi_{p1} X_p$$

that has the largest variance, subject to the constraint that $\sum_{j=1}^{p} \phi_{j1}^2 = 1$. The ϕs are the loadings of the principal components. For data set with p variables, there can be up to p principal components.

Math Behind PCs

The second principal component is the linear combination that has maximal variance out of all linear combinations that are uncorrelated with the first principal component (i.e. orthogonal to the first eigenvector).

Math Behind PCs

There are a number of matrix decomposition techniques that can be used to calculate PCs for any given data set, with singular value decomposition of the data matrix and eigendecomposition of the covariance matrix being two notable ones. The math behind these is beyond the scope of the class.

Revisiting our exercise

- Imagine applying PCA to the O-levels – what kinds of principal components might they reveal?

An O-level example

	PC1	PC2
English	0.41	0.02
E. Math	0.41	0.45
Physics	0.41	0.48
Chemistry	0.41	0.42
Literature	0.40	−0.46
History	0.40	−0.44

Applied PCA

Steps behind PCA

1. Choose variables of interest.
2. Normalize variables (e.g. scale it to have a mean of zero, and SD of 1).
3. Conduct PCA.
4. Choose the appropriate amount of principal components.
5. Interpret the PC loadings.
6. Use PCs in other analysis, whether it's visualizations or regressions.

Output 1: Variance Explained

Five GSS variables: income, occupational prestige, education, number of children, and age. Each is scaled to mean 0 and SD 1.

```
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

cols = ["income_cont", "prestg10", "educ", "biokids", "age"]
X = StandardScaler().fit_transform(df[cols])
pca = PCA().fit(X)

prop = pca.explained_variance_ratio_
print(pd.DataFrame({
    "Standard deviation": pca.explained_variance_ ** 0.5,
    "Proportion of variance": prop,
    "Cumulative proportion": prop.cumsum(),
}, index=["PC1", "PC2", "PC3", "PC4", "PC5"]).T.round(3))
```

```
                          PC1    PC2    PC3    PC4    PC5
Standard deviation      1.368  1.127  0.882  0.781  0.689
Proportion of variance  0.374  0.254  0.155  0.122  0.095
Cumulative proportion   0.374  0.628  0.783  0.905  1.000
```

Output 2: PC Loadings

Loadings are the weights in $Z = \phi_1 X_1 + \cdots + \phi_p X_p$. In `sklearn`, each row of `pca.components_` is one PC. The table below is that matrix transposed, so each column is a PC.

```
loadings = pd.DataFrame(
    pca.components_.T,
    index=cols,
    columns=["PC1", "PC2", "PC3", "PC4", "PC5"],
)
print(loadings.round(3))
```

```
                PC1    PC2    PC3    PC4    PC5
income_cont   0.524  0.009 -0.517  0.676 -0.030
prestg10      0.592  0.095  0.042 -0.457 -0.656
educ          0.594 -0.112  0.209 -0.268  0.721
biokids      -0.098  0.710 -0.560 -0.351  0.223
age           0.109  0.689  0.612  0.373 -0.022
```

How many PCs are Needed?

The intuition is usually the smallest amount of principal components required to get a “good understanding" of the data.
No definitive answer, but consider the scree plot.

Scree Plot

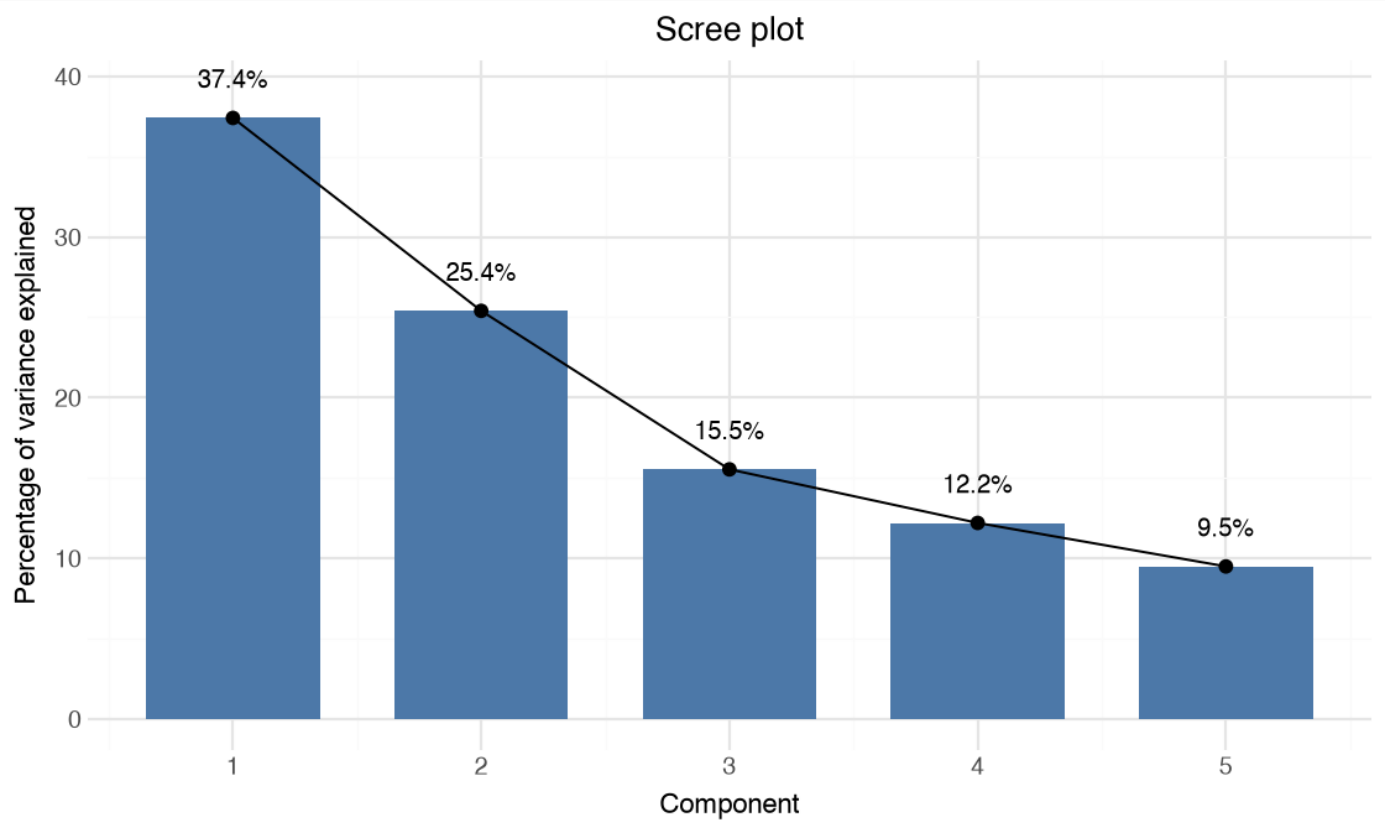
```
from plotnine import (
    ggplot, aes, geom_col, geom_line, geom_point, labs
)

scree = pd.DataFrame({
    "component": range(1, 6),
    "percent": pca.explained_variance_ratio_ * 100,
})

(ggplot(scree, aes("component", "percent"))
 + geom_col(fill="#4C78A8", width=0.7)
 + geom_line(aes(group=1))
 + geom_point()
 + labs(x="Component", y="Percentage of variance explained")
)
```

Scree Plot

Figure 1: Look for "elbows" in the scree plot.



Example: Daenekindt & Roose's (2013) work on social mobility and film taste.

Daenekindt & Roose (2013)

Does a change in education change how a person watches films, or does the taste from the parental home stay?

Daenekindt & Roose (2013)

- Sample: Cultural Participation in Flanders, 2003–2004. 2,849 adults, interviewed face to face. Response rate 61 per cent.
- The questions take the form "to what extent do you think that a film should . . . ?" Eight items, each on a 7-point scale from never important to always important.

Daenekindt & Roose (2013)

- Eight items are too many outcomes. They reduce them to three dispositions.
- The tool is an exploratory factor analysis with a Promax rotation. Three factors have an eigenvalue above 1, so three dispositions are kept.
- Read the pattern matrix as a loadings table. A large loading means that item defines that disposition.

Daenekindt & Roose (2013)

Table 1: Pattern matrix for aesthetic dispositions

	Innovative	Emotional	Action
Original in outline or style	0.740	0.011	−0.051
Special or creative camera work	0.730	−0.073	0.080
Innovative in directing	0.773	0.062	−0.038
Be moving	0.017	0.752	−0.048
Be romantic	−0.090	0.778	−0.005
Characters you can relate to	0.141	0.470	0.110
Violent scenes	−0.003	0.103	0.671
Action and adventure	−0.015	−0.078	0.743

Interpreting the loadings

- "Be innovative in directing" loads 0.773 on the first factor and near zero on the other two. That item defines the innovative disposition.
- "Contain action and adventure" loads 0.743 on the third factor. That item defines the action disposition.
- "Contain characters you can relate to" loads 0.470 on the emotional factor, the weakest of the eight, with a smaller loading of 0.141 on the innovative factor. They still place it with the emotional disposition.

What they do with the scores

Each respondent receives a score on each disposition. Those three scores are the dependent variables.

- On the innovative disposition, the least educated score lower (-0.215) than the higher educated (0.209).
- For the upwardly mobile, the weight on the parents' education is 0.285. For the downwardly mobile it is $0.285 + 0.615$. The higher-status way of watching is the one that stays.
- On the emotional and action dispositions, origin and destination both matter. Women score higher on the emotional disposition (0.581); men score higher on action.

Thanks for listening :)